

Vitamin D Deficiency & Rickets in Children

Clinical Summary for Physicians — based on UpToDate: Misra M, et al. "Vitamin D insufficiency and deficiency in children and adolescents" (Jun 2020)

1. DEFINITION & METABOLISM

Vitamin D is a prohormone essential for calcium/phosphorus homeostasis and bone mineralization. Severe deficiency causes **rickets** (growing children — failure of growth-plate/bone mineralization) and **osteomalacia** (any age — impaired mineralization of already-formed bone). Mild/chronic deficiency is linked to reduced bone mineral density even without overt rickets.

Key Forms of Vitamin D

- **Vitamin D3 (Cholecalciferol)** — made in skin by UVB; found in animal products/supplements.
- **Vitamin D2 (Ergocalciferol)** — from plant sources; found in most supplements.
- **Calcidiol (25-OH-D)** — storage form made in the **liver**; the best lab marker of vitamin D status.
- **Calcitriol (1,25-(OH)₂-D)** — active hormone made in the **kidney**; drives intestinal calcium absorption, bone resorption, and renal calcium/phosphate handling.

Pathway: Skin (UVB) or Diet → Cholecalciferol/Ergocalciferol → Liver (25-OH-D, storage) → Kidney (1,25-(OH)₂-D, active) → ↑ gut Ca absorption, ↑ bone resorption, ↓ renal Ca/PO₄ loss.

2. DIAGNOSTIC CLASSIFICATION (based on serum 25-OH-D)

Category	25-OH-D (ng/mL)	25-OH-D (nmol/L)
Sufficiency	20 – 100	50 – 250
Insufficiency	12 – 20	30 – 50
Deficiency	< 12	< 30

Recommended daily intake: Infants (0–12 mo) = 400 IU/day; Children/adolescents (1–18 yr) = 600 IU/day (higher in high-risk groups, up to 2000–6000 IU/day).

3. STAGES OF VITAMIN D DEFICIENCY (Biochemical Progression)

Stage	Ca	PO4	ALP	PTH	25-OH-D	1,25-(OH)2-D	X-ray
Early	N/↓	N/↓	↑	↑	↓	N	Reduced mineralization
Moderate	N/↓	↓	↑↑	↑↑	↓↓	↑	Rachitic changes +
Severe	↓↓	↓↓	↑↑↑	↑↑↑	↓↓↓	↑/N/↓	Rachitic changes ++

Pathophysiology in one line: ↓25-OH-D → ↓gut Ca/PO4 absorption → ↑PTH (mobilizes bone Ca, wastes PO4 renally) → low Ca×PO4 product → defective mineralization → rickets/osteomalacia.

4. CLINICAL MANIFESTATIONS

Most patients are asymptomatic. When symptomatic:

Skeletal (Rickets — growing children)

- Delayed fontanelle closure, craniotables, frontal bossing
- Widened wrists/ankles, rachitic rosary (costochondral prominence)
- Bow legs (genu varum) or knock knees (genu valgum)
- Motor delay, poor growth, bone pain/irritability
- X-ray: metaphyseal widening/splaying/cupping/fraying, loss of provisional calcification zone

Osteomalacia (older adolescents/adults — growth complete)

- Diffuse or localized bone and muscle pain/weakness (no deformity, since growth plates are fused)

Severe / Hypocalcemic Manifestations

- Muscle weakness, difficulty standing/walking (from hypophosphatemia)
- Tetany, seizures, apnea, stridor, wheezing, hypotonia, hyperreflexia (from severe hypocalcemia — most common in infancy and adolescence, periods of rapid growth)

5. EPIDEMIOLOGY

- US pediatric prevalence of deficiency/insufficiency (25-OH-D <20 ng/mL): ~15%; severe deficiency (<10 ng/mL) in 1–2%.
- Europe: ~45% of children/adolescents have insufficiency or deficiency.
- Highest rates of rickets: Black children > Asian Indian children > White children — tracking with skin pigmentation.
- Black non-Hispanic adolescents have ~20× the risk of low 25-OH-D vs White non-Hispanic adolescents (NHANES III); risk higher in females.
- Deficiency is far more common at end of winter than end of summer; more common at latitudes beyond 40°.

6. ETIOLOGY & RISK FACTORS

Perinatal / Infant Risk Factors

- **Maternal vitamin D deficiency** during pregnancy (crosses placenta; critical in 3rd trimester for fetal skeletal calcification)
- **Prematurity** — less time for transplacental transfer
- **Exclusive breastfeeding beyond 3–6 months** without supplementation — breast milk vitamin D content is low (major cause: >95% of US rickets cases were in breastfed infants)
- Dark maternal/infant skin pigmentation

Common Postnatal Risk Factors (Older Children)

- Dark skin pigmentation (melanin blocks UVB synthesis; darker skin needs 3–10× more sun exposure)
- Low dietary intake — limited dairy, vegan/unusual diets, poor fortified-food intake
- High latitude residence, winter season, limited sun exposure, consistent sunscreen use, staying mostly indoors
- **Obesity** — vitamin D sequestered in fat tissue → higher requirement

Medications & Medical Conditions

- Anticonvulsants, antiretrovirals (increase 25-OH-D catabolism)
- Chronic glucocorticoids (impair intestinal Ca absorption)
- Ketoconazole/antifungals (block 1-hydroxylation)
- Malabsorptive conditions: celiac disease, IBD, cystic fibrosis (pancreatic insufficiency), cholestatic liver disease, post gut-resection/bariatric surgery
- Chronic kidney disease, severe hepatic dysfunction

Genetic Causes (rare)

- 25-hydroxylase deficiency (CYP2R1 mutation)
- 1-alpha-hydroxylase deficiency (CYP27B1 mutation) — low 1,25-(OH)₂-D, normal 25-OH-D
- Hereditary vitamin D resistance (VDR gene mutation) — alopecia + normal/high 25-OH-D and 1,25-(OH)₂-D

Note: In nephrotic syndrome, low total 25-OH-D may reflect low vitamin D-binding protein (free/bioavailable 25-OH-D may be normal) — not true deficiency.

7. WHOM & HOW TO TEST (Screening)

Screen only at-risk groups (not the general population):

- Exclusively breastfed/premature infants not reliably taking vitamin D supplements
- Dark-skinned children at higher latitudes (screen in winter/spring when levels are lowest)
- Children with low dietary vitamin D intake and no supplementation
- Infants/children with unexplained poor growth, motor delay, or irritability
- Children on anticonvulsants, antiretrovirals, or chronic glucocorticoids
- Chronic malabsorptive disease, low bone density conditions, CKD, or severe liver disease
- Pregnant or lactating adolescents
- Markedly elevated alkaline phosphatase for age (>500 IU/L in neonates; >1000 IU/L in children <9 yr)

(Screening in obesity alone is controversial/optional.)

8. INVESTIGATIONS

- **Serum 25-OH-D** — the test of choice for vitamin D status (best reflects total body stores; half-life 2–3 weeks)
- 1,25-(OH) $_2$ -D is **not** useful for screening (short half-life, PTH-driven fluctuations, may be normal/high even in deficiency)
- If 25-OH-D <20 ng/mL: add serum calcium, phosphorus, alkaline phosphatase, and PTH
- Radiographs (wrist/knee) if age <3 years or high clinical suspicion of rickets
- Classify rickets as calcipenic (hypocalcemic) vs hypophosphatemic based on labs

9. DIAGNOSTIC THRESHOLDS (Recap)

Sufficiency ≥ 20 ng/mL (50 nmol/L) · Insufficiency 12–20 ng/mL (30–50 nmol/L) · Deficiency <12 ng/mL (<30 nmol/L).
Radiologic rickets/low bone density appear below ~ 16 – 18 ng/mL; ALP begins rising below ~ 20 ng/mL.

10. MANAGEMENT (Treatment)

Treat if 25-OH-D <20 ng/mL (50 nmol/L) or clinical/radiologic rickets. Either D2 or D3 may be used.

Group	Treatment dose	Duration → Maintenance
Infants <12 mo	2000 IU/day	6–12 wk → then ≥400 IU/day
Children ≥12 mo (deficiency, no rickets)	2000 IU/day (or 50,000 IU once weekly ×6 wk)	6–12 wk → then 600–1000 IU/day
Established rickets, 12 mo–12 yr	3000–6000 IU/day	12 wk → then maintenance
Established rickets, ≥12 yr	6000 IU/day	12 wk → then maintenance
Obesity / malabsorption / interacting medications	2–3× standard dose (up to 6000 IU/day)	Higher maintenance dose

Calcium Co-supplementation

- Give if elevated PTH or rickets present — prevents "hungry bone syndrome" hypocalcemia during repletion
- Elemental calcium 30–75 mg/kg/day (divided 2–3×/day) for 2–4 weeks, or per Global Consensus: 500 mg/day elemental calcium
- Severe symptomatic hypocalcemia (seizures/tetany): IV calcium gluconate 10–20 mg/kg elemental Ca (max 540 mg/dose), slow IV

Monitoring / Follow-up

- Check calcium after 1–2 months in all infants and anyone on >2000 IU/day (risk of hypercalcemia, esp. CYP24A1 mutations)
- Uncomplicated low 25-OH-D: recheck at 2–3 months
- Elevated ALP/PTH without rickets: recheck at 6–8 weeks, then periodically, then annually
- Rickets: close follow-up for radiographic healing and normalization of labs

11. PREVENTION

- **All exclusively breastfed infants:** 400 IU/day vitamin D starting within days of birth, continued until weaned onto ≥1 L/day of vitamin D-fortified formula or milk
- Formula-fed infants consuming <1 L/day of fortified formula also need supplementation
- Alternative: high-dose maternal supplementation (4000–6400 IU/day) can adequately enrich breast milk
- Pregnant women: ensure adequate intake (some data suggest >600 IU/day needed, especially in dark-skinned women)
- Older children: encourage vitamin D-fortified dairy/foods; routine sun exposure is NOT recommended as a prevention strategy (skin cancer risk)

12. KEY TAKE-HOME POINTS

- Rickets = pediatric mineralization failure at growth plates; osteomalacia = mineralization defect after growth plates fuse.
- 25-OH-D is the correct screening test — not 1,25-(OH)₂-D.
- Screen only high-risk groups; treat if 25-OH-D <20 ng/mL or rickets present.
- Give calcium alongside vitamin D when PTH is elevated or rickets is present, to avoid hungry-bone hypocalcemia.
- Prevention (400 IU/day for breastfed infants; 600 IU/day for children) is simple, safe, and effective.

Source: Misra M. "Vitamin D insufficiency and deficiency in children and adolescents." UpToDate, literature review current through May 2020, topic last updated Jun 11, 2020. This is a condensed educational summary — refer to the original topic review and local guidelines for full clinical detail and dosing verification.